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A Novel Strategy to Produce CAR- $\gamma\delta$ T Cells via Site-Directed Gene Integration by a Combination of CRISPR/Cas9 and AAV

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ABSTRACT

Chimeric antigen receptor (CAR)- $\alpha\beta$ T cells are commonly employed in tumor therapy but hindered by some limitations. $\gamma\delta$ T cells are promising substrates for CAR therapy for their major histocompatibility complex (MHC)-unrestricted recognition manner and innate immune function. Here, we established a novel method for generating CAR- $\gamma\delta$ T cells. We utilized the clustered regularly interspaced palindromic repeats (CRISPR)/CRISPR-associated protein 9 (Cas9) (CRISPR/Cas9) method to interrupt the *TCR delta chain constant region (TRDC)* sequence, followed by the site-directed insertion of the CAR sequence into the *TRDC* locus via homologous complementation mediated by adeno-associated virus (AAV) gene delivery. We optimized electroporation parameters for Cas9/ribonucleoproteins (RNP) delivery and infection conditions for CAR-gene carrying AAV in $\gamma\delta$ T cells. These optimizations facilitated efficient TCR knockout and site-directed CAR insertion, ultimately yielding functional CAR- $\gamma\delta$ T cells. In vitro experiments demonstrated that these newly prepared CAR- $\gamma\delta$ T cells could stimulate cytokine production, kill tumor cells as well as exhibit robust proliferative potential and memory-like phenotype. These state-of-the-art CAR- $\gamma\delta$ T cells could reduce tumor burden and extend the survival period of tumor-bearing mice.

1 | Introduction

Chimeric antigen receptor (CAR)-T cell therapy has emerged as a transformative treatment modality for cancer. Conventional CAR-T cell therapy involves the genetic modification of autologous $\alpha\beta$ T cells to express a synthetic receptor that specifically recognizes tumor-associated antigens (TAAs) in a major histocompatibility complex (MHC)-unrestricted manner, thereby triggering T-cell activation, cytotoxicity, cytokine secretion, and proliferation. Since 2017, the U.S. FDA has approved six CAR- $\alpha\beta$

T-cell therapy products for clinical use: Yescarta, Kymriah, Tecartus, Abecma, Breyanzi, and Carvykti. Currently, CAR- $\alpha\beta$ T-cell therapy stands as a leading therapeutic strategy in oncology. However, its application is constrained by critical limitations, including suboptimal efficacy in solid tumors, immune evasion driven by the downregulation of TAAs expression in neoplastic cells, and off-target toxicities resulting in neuronal damage due to the aberrant expression of CD19 on mural cells. These challenges underscore the urgent need to optimize the therapeutic approach to enhance both safety and efficacy [1–4].

Ruoyu Dong and Yixi Zhang contributed equally to this study.

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Beyond $\alpha\beta$ T cells, $\gamma\delta$ T cells have garnered increasing attention owing to their unique biological characteristics. First, $\gamma\delta$ T cells recognize antigens in an MHC-unrestricted manner (e.g., BTN molecules, MICA/B, and EphA2) [5], which avoids the risk of graft-versus-host disease (GVHD). Hence, $\gamma\delta$ T cells are ideal candidates for developing “off-the-shelf” CAR-T therapy. Second, as a specialized subset of innate-like T lymphocytes, $\gamma\delta$ T cells possess the capacity to express natural killer (NK) cell receptors, thereby mediating innate immune responses. This dual recognition system may strengthen the anti-tumor effect in addition to CAR mediated cytotoxicity. For instance, tumor cells in the tumor microenvironment often downregulate molecules involved in immune recognition, such as MHC-I and TAA, to evade $\alpha\beta$ T-cell-induced antitumor response. However, the dual recognition system of $\gamma\delta$ T cells may be beneficial to eliminate tumor cells [6, 7]. As of now, CAR- $\gamma\delta$ T cell therapies have entered clinical trials for various malignancies, including B-cell lymphoma, renal cell carcinoma and CD7-positive T-cell derived malignancies (NCT04735471, NCT06480565, NCT04702841).

Currently, CAR- $\gamma\delta$ T cells are commonly manufactured using lentiviral (LV) or gamma retroviral (γ RV) vectors. These methods carry risks of malignant transformation due to insertional mutagenesis and unstable anti-tumor activity [8]. For example, patients with X-linked severe combined immunodeficiency (X-SCID) developed T-cell leukemia following γ RV-based gene therapy, likely due to the integration of the therapeutic gene into oncogenes. Recent data also indicate that patients treated with LV-based gene therapy products, such as Zynteglo and Skysona, have subsequently developed hematologic malignancies [9], highlighting the unmet need for further safety validation.

To circumvent the disadvantages of conventional CAR- $\alpha\beta$ T cell manufacturing methods, Justin et al. proposed a combination of Cas9/ribonucleoproteins (RNP) (Cas9/RNP) and AAV gene delivery. In their approach, they simultaneously knocked out TCR and performed site-directed CAR gene knock-in at the *T cell receptor alpha constant (TRAC)* locus. AAV is a non-pathogenic parvovirus with a single-stranded DNA genome (maximum packaging capacity ~4.7 kb) enclosed in a capsid. As a versatile gene delivery tool, it has been widely applied to transduce diverse mammalian tissues and cell types, providing long-term, stable transgene expression. Among natural AAV serotypes, AAV6 exhibits robust tropism for primary human CD4⁺ and CD8⁺ T cells. Specifically, AAV6 mediates homology-directed repair (HDR)-based gene editing in activated primary human $\alpha\beta$ T cells with efficiencies reaching up to 50% [10]. By this way, the endogenous promoter of *TRAC*, rather than random endogenous or exogenous promoters, drives CAR gene expression, resulting in superior anti-tumor function [11]. This research inspired the improvement of CAR-T cell manufacturing system, prompting us to investigate whether the combination of Cas9/RNP and AAV6 gene delivery is feasible for generating CAR- $\gamma\delta$ T cells.

Here, we developed a Cas9/RNP-AAV protocol specifically optimized for $\gamma\delta$ T cells. We employed the Cas9/RNP method to precisely cleave the specific site of the *TCR delta chain constant region (TRDC)* sequence and utilized AAV to deliver the CAR gene sequence under the control of endogenous TCR δ regulatory elements. Consequently, we achieved approximately 90%

knockout (KO) efficiency and 20% targeted insertion efficiency in $\gamma\delta$ T cells. This protocol successfully enabled the production of CAR- $\gamma\delta$ T cells, which demonstrate a certain antitumor potency both in vitro and in vivo.

2 | Results

2.1 | Expansion of $\gamma\delta$ T Cells From PBMCs Utilizing Zoledronic Acid (ZOL) and IL2

Currently, the combination of ZOL and IL2 (ZOL/IL2) is commonly used to induce peripheral blood mononuclear cells (PBMCs) into $\gamma\delta$ T cells [12, 13]. Given that IL7 and IL15 have been reported to support the generation of CAR-T cells with a memory-like phenotype which usually exhibits enhanced proliferation and antitumor potential [14, 15], we also performed cell-expansion experiments of $\gamma\delta$ T cells using PBMCs induced by the combination of ZOL and IL7 or IL15, respectively. We found that after 19 days of culture, all conditions except ZOL combined with IL7 alone could induce PBMC-derived $\gamma\delta$ T cells with a purity exceeding 70% while maintaining rapid proliferation. Among them, PBMCs stimulated by ZOL/IL2 or ZOL combined with IL7 and IL15 (ZOL/IL7/IL15) achieved the highest production of $\gamma\delta$ T cells, with cell number increasing approximately 140-fold (Figure 1A,B). Furthermore, we compared the exhaustion status of cells under these two conditions and observed that $\gamma\delta$ T cells stimulated by ZOL/IL2 exhibited lower expression levels of exhaustion molecules (PD1, TIM3, LAG3) and a higher proportion of cells with less exhausted phenotype (Figure 1C). Therefore, we selected ZOL/IL2 for the induction of the PBMC-derived $\gamma\delta$ T cells in the subsequent experiments.

2.2 | Electroporation of Cas9/Ribonucleoproteins (RNP) Enables Efficient TCR Knockout in Human $\gamma\delta$ T Cells

To achieve efficient and rapid gene knockout (KO), the method of delivering Cas9/RNP, complexes of recombinant Cas9 with sgRNA, is widely utilized [16, 17]. RNP can be prepared using chemically modified synthetic target gene-specific CRISPR RNAs (crRNA) and tracer RNAs (tracrRNAs), or synthesized sgRNA. Both can achieve 90% efficient targeted knockout in $\alpha\beta$ T cells [18]. Due to the simplicity of operation, we chose to use directly synthesized sgRNA combined with Cas9 protein to prepare RNP.

To optimize Cas9 methodology, we screened a series of potential guide RNA (gRNA) targeting the *TRDC* locus by evaluating the frequencies of insertions or deletions (INDEL) by tracking of INDELs by decomposition (TIDE) analysis in K562 cell line and the efficacy of TCR expression reduction in $\gamma\delta$ T cells using different gRNAs. Finally, we selected gRNA1 for further studies, which achieved the highest INDEL frequency and a TCR KO rate of ~80% (Figure 2A, Figure S1A).

Considering that the Cas9/RNP electroporation parameters commonly used for human or mouse $\alpha\beta$ T cells might not be optimal for human $\gamma\delta$ T cells, we developed a Cas9/RNP transfection protocol specific for human $\gamma\delta$ T cells. Using TCR as a target, we

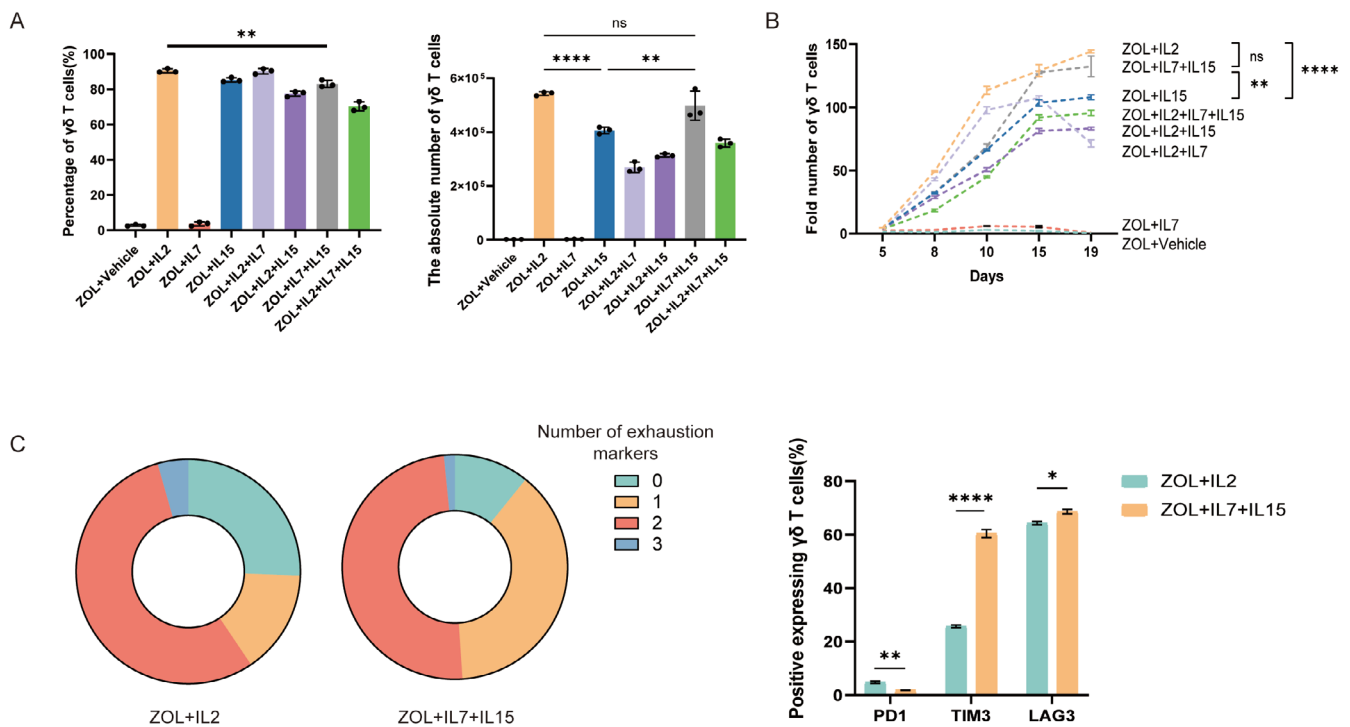


FIGURE 1 | Stimulation and expansion of $\gamma\delta$ T cells from PBMCs. (A, B) $\gamma\delta$ T cells were induced by stimulation with ZOL and different combinations of cytokines (IL2, IL7, IL15) in an initial number of 2×10^5 PBMCs. The purity (left) and absolute number (right) of $\gamma\delta$ T cells obtained after stimulating PBMCs for 19 days (A) and the proliferation of $\gamma\delta$ T cells after stimulation (B) are shown. (C) Percentage of $\gamma\delta$ T cells expressing exhaustion markers, quantified by FACS at Day 19. Data are presented as mean $\pm$ SD ($n = 3$, technical triplicates). p -values are determined by t -tests or ANOVA. * $p < 0.05$; ** $p < 0.01$; **** $p < 0.0001$; ns, not significant ($p > 0.05$).

first determined the optimal electroporation program to achieve the highest efficacy for TCR KO. Based on the results presented in Figure 1, $\gamma\delta$ T cells maintain high cellular viability and proliferation rates even after prolonged treatment with ZOL. Thus, we transfected 2million PBMCs cultured and expanded in medium supplemented with ZOL and IL2 for 5days (D5) or for 10days (D10), keeping an amount of Cas9 constant at $15 \mu\text{g}$ and a ratio of gRNA/Cas9 of 2.5, as referenced from established protocol for $\alpha\beta$ T cells [19]. Using a systematic optimization approach testing different electroporation programs (X001, T007, U014, T023; Lonza IIB), we found human PBMC-derived $\gamma\delta$ T cells receiving X001 program on D5, or X001 program on D10 showed higher rates of KO efficiency, cell viability, and cell recovery (Figure 2B,C). Key parameters of the Cas9/RNP complex—including Cas9 protein dosage and the gRNA to Cas9 molar ratio—have been shown to modulate CRISPR/Cas9-mediated KO efficiency in various cell types [18]. To improve KO efficiency in $\gamma\delta$ T cells, we attempted to optimize the conditions of Cas9/RNP, including appropriate Cas9 amount, ratio of gRNA to Cas9, and cell number titration, in a convenient and easy-to-practice setting. First, we used $15 \mu\text{g}$ of Cas9 to transfect 2million PBMCs in a reaction using electroporation program X001. The KO efficiency increased as the gRNA/Cas9 ratio increased, which finally reached $>80\%$ when the ratio was 4 and showed the best survival capacity (Figure 2D, Figure S1B). When keeping gRNA/Cas9 ratio of 4, $10 \mu\text{g}$ of Cas9 was sufficient to provide the best KO efficacy and good ability of cell survival (Figure 2E, Figure S1C), whereas further increase in Cas9 amount could not cause more improvement (Figure 2E). Besides, we also explored the effect of the cell number in a transfection reaction on KO efficiency. We observed that 1–2million cells in one reaction

could achieve the highest KO rate. However, further increase in transfected cell number led to reduced KO efficiency (Figure 2F). As a further validation, this protocol was also applied to target HLA in human $\gamma\delta$ T cells, achieving approximately 70% KO efficiency of HLA (Figure S1D). Collectively, an optimal KO system for disrupting the TCR in $\gamma\delta$ T cells has been successfully established.

2.3 | Site-Directed CAR Konckin in Human $\gamma\delta$ T Cells by AAV6 Transduction

Following Cas9-introduced double-stranded break (DSB) formation, AAV was widely used to deliver donor templates for primary immune cells to provide the targeted DNA template. Compared to other capsid serotypes, AAV6 was reported to efficiently transduce $\alpha\beta$ T cells [10]. We therefore investigated whether AAV6 could achieve high transduction efficiency in $\gamma\delta$ T cells. AAV6 carrying a reporter protein gene (enhanced green fluorescence protein, EGFP) under the control of CMV promoter was used to transduce $\gamma\delta$ T cells, and achieved a high transduction efficiency of 90% (Figure S1E). To precisely insert CAR gene at the TRDC site, we designed an AAV6 vector carrying left and right homology arms (HA) that flanked the gRNA targeting sequences mentioned above, then packaged this construct into the AAV6 viral capsid (Figure 3A). Previous reports indicated that HA length was an important variable for efficient HDR-mediated DNA repair, with maximum gene editing efficacy achievable using short 400bp HA in human cell lines and primary T cells [20]. Thus, we recommended the plasmid donor templates with 400bp HA for $\gamma\delta$ T cells. Other details of the AAV6 vector

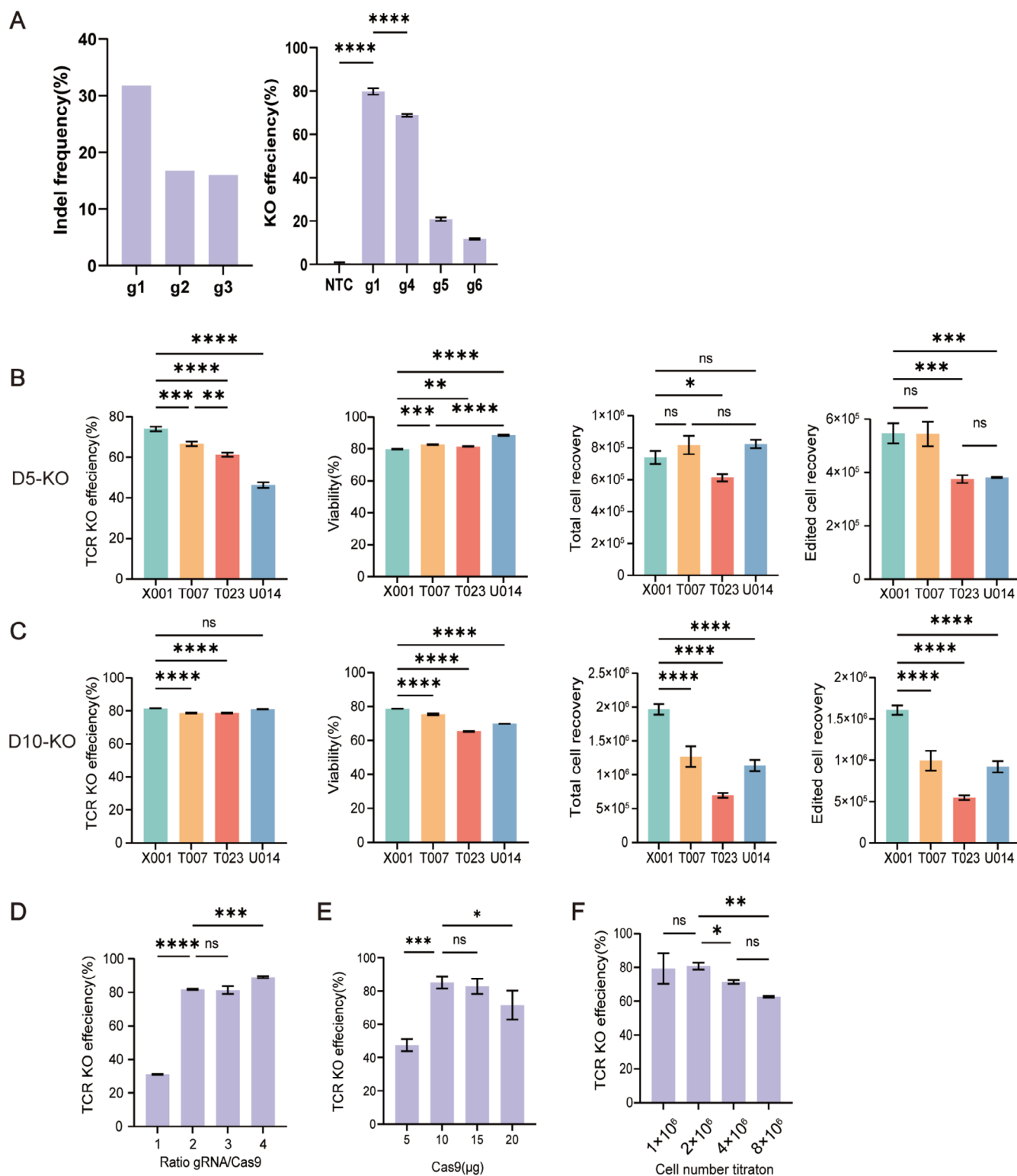


FIGURE 2 | TCR KO in human $\gamma\delta$ T cells. (A) The functional gRNA screening via the INDEL frequency in K562 cell line using tracking of INDELs by decomposition (TIDE) analysis (left) and actual loss of TCR expression in $\gamma\delta$ T cells (right). Data are presented as mean $\pm$ SD ($n = 3$, technical triplicates). (B, C) TCR KO efficiency, cell viability, total cell recovery, and edited cell recovery (TCR-KO cells) measured 4 days after electroporation of Cas9/RNP by different programs (X001, T007, T023, U014; Lonza IIB) targeting *TRDC* locus using PBMCs cultured for 5 days (D5) (B) or for 10 days (D10) (C). Data are presented as mean $\pm$ SD ($n = 3$, technical triplicates). (D-F) TCR KO efficiency measured 4 days after electroporation of Cas9/RNP targeting *TRDC* locus, and titration of gRNA to Cas9 ratio (D), the Cas9 amount (E), transfected cell number (F). Data are presented as mean $\pm$ SD ($n = 3$, technical triplicates). p -values are determined by ANOVA. * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$; **** $p < 0.0001$; ns, not significant ($p > 0.05$).

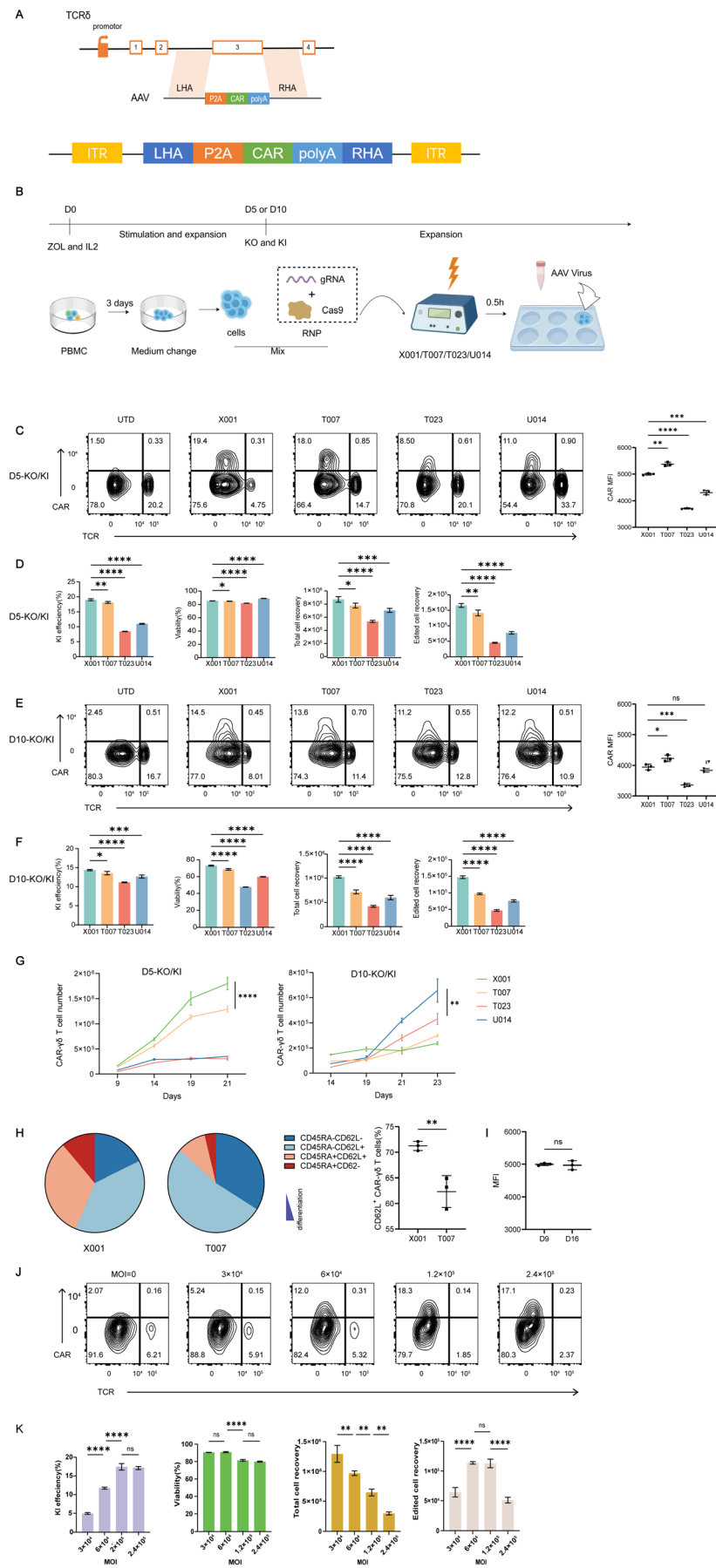


FIGURE 3 | Legend on next page.

FIGURE 3 | Site-directed CAR gene delivery in $\gamma\delta$ T cells. (A) Descriptive graphic illustrating the integration by homology-directed repair of the CAR sequence in the exon3 of the *TRDC* locus following a double-stranded break introduction by CRISPR/Cas9 (upper) and the specific rAAV6 construct containing the CAR cassette flanked by 400 bp long homology arms (lower). (B) Specific process flowchart for preparing CAR- $\gamma\delta$ T cells. PBMCs were cultured for 5 or 10 days in medium supplemented with ZOL and IL2 and then transduced with Cas9/RNP via different electroporation protocols, followed by infection with AAV6 carrying the CAR gene at different MOIs 0.5 h after electroporation. (C–F) CAR-KI efficiency was assessed at 4 days after electroporation of Cas9/RNP using different programs (X001, T007, T023, U014; Lonza IIB) to target the *TRDC* locus in PBMCs expanded for 5 or 10 days (D5 or D10). Representative flow plots and analysis of MFI of CAR expression (C, E) and the analysis of CAR-KI efficiency, cell viability, recovery potential of total cells, and edited cells (TCR-KO and CAR-KI $\gamma\delta$ T cells) (D, F) are shown. Data are presented as mean $\pm$ SD ($n=3$, technical triplicates). (G) Proliferative potential of CAR- $\gamma\delta$ T cells (D5/10, X001/T007/T023/U014). Data are presented as mean $\pm$ SD ($n=3$, technical triplicates). (H) Phenotype (left) and the CD62 expression level (right) of CAR- $\gamma\delta$ T cells (D5, X001/T007). Data are presented as mean $\pm$ SD ($n=3$, technical triplicates). (I) The MFI of CAR expression of CAR- $\gamma\delta$ T cells cultured at Day 9 (D9) and Day 16 (D16). Data are presented as mean $\pm$ SD ($n=3$, technical triplicates). (J, K) CAR-KI efficacy was assessed at different MOIs. Representative flow plots (J) and the analysis of CAR-KI efficiency, cell viability, recovery potential of total cells and edited cells (TCR-KO and CAR-KI $\gamma\delta$ T cells) (K) were shown. Data are presented as mean $\pm$ SD ($n=3$, technical triplicates). p -values are determined by t -tests or ANOVA. * $p<0.05$; ** $p<0.01$; *** $p<0.001$; **** $p<0.0001$; ns, not significant ($p>0.05$).

design have been well described [21]. We subsequently employed PBMC-derived $\gamma\delta$ T cells stimulated for 5 or 10 days, electroporated with Cas9/RNP targeting *TRDC* locus, followed by AAV6 transduction to generate CAR- $\gamma\delta$ T cells (marked as CAR- $\gamma\delta$ T cells-D5/10, X001/T007/T023/U014) (Figure 3B). We evaluated KI efficiency, mean fluorescence intensity (MFI), cell viability, recovery ability of total cells and edited cells (TCR negative and CAR positive cells) 4 days after manufacture. Based on these assessment indicators, CAR- $\gamma\delta$ T cells-(D5, X001) and CAR- $\gamma\delta$ T cells-(D10, X001) exhibited the highest rates of KI efficiency, cell viability and the highest number of CAR- $\gamma\delta$ T cells, as well as relatively high CAR gene expression levels (Figure 3C–F). We further expanded these cells for ~2 weeks and assessed a long-term proliferative potential of CAR- $\gamma\delta$ T cells. Noteworthy, although the CAR- $\gamma\delta$ T cells-(D10, U014) displayed a high-speed of proliferation in the later period of cell culture, the cell numbers of the CAR- $\gamma\delta$ T cells-(D5, X001) and CAR- $\gamma\delta$ T cells-(D5, T007) were about 2–3 times higher than those of CAR- $\gamma\delta$ T cells-(D10, U014) (Figure 3G). Furthermore, compared to the CAR- $\gamma\delta$ T cells-(D5, T007), CAR- $\gamma\delta$ T cells-(D5, X001) exhibited a higher frequency of cells with memory phenotype with a higher expression level of CD62L, maintaining a stable CAR expression level over an extended culture time (Figure 3H,I). Taken together, we finally determined the optimal protocol for generating CAR- $\gamma\delta$ T cells, where PBMCs stimulated by ZOL/IL2 for 5 days were transfected with Cas9/RNP using electroporation program X001, then followed by site-directed CAR insertion via AAV6 gene delivery. The optimal transduction efficiency at D5 can be attributed to its alignment with established CAR- $\gamma\delta$ T cell protocols, where genetic modification is conventionally performed at 2–5 days post ZOL stimulation [22, 23]. Finally, we observed that the KI efficiency was positively correlated with multiplicity of infection (MOI) when MOI was below 1.2×10^5 , whereas a further increase of MOI was not meaningful as high MOI resulted in a reduced viability and impaired cell recovery potential (Figure 3J,K).

2.4 | CAR- $\gamma\delta$ T Cells Exhibit Anti-Tumor Function In Vitro

Using the protocol described above, we generated CD19-targeting CAR- $\gamma\delta$ T cells. Then we evaluated their anti-tumor function

against TAA (CD19)-positive Nalm6, a human pre-B ALL cell line. CD19-targeting CAR- $\gamma\delta$ T cells showed a pronounced ability to secrete interferon gamma (INF γ) and tumor necrosis factor alpha (TNF α) when co-cultured with CD19-positive Nalm6 for 18 h, whereas untransduced (UTD) $\gamma\delta$ T cells did not (Figure 4A,B). Interestingly, CAR- $\gamma\delta$ T cells contained higher percentages of naïve/central memory subsets (CD45RA⁺CD62L⁺ and CD45RA⁺CD62L⁺) than UTD cells either in resting state or stimulated conditions (Figure 4C,D). This naïve/central memory phenotype was associated with improved anti-tumor function [24, 25], indicating the potential of the ideal anti-tumor activity of CAR- $\gamma\delta$ T cells even after stimulation in the tumor microenvironment. We further evaluated the proliferation capacity via carboxyfluorescein diacetate succinimidyl ester (CFSE). CAR- $\gamma\delta$ T cells displayed significantly stronger proliferation capacity in response to antigen stimulation compared with the UTD group (Figure 4E). Most importantly, although the UTD $\gamma\delta$ T cells exerted slight cytotoxicity towards CD19 positive tumor cells, CAR- $\gamma\delta$ T cells had a more robust ability to lyse different tumor cells even at a low effector/target ratio (E/T) (Figure 4F). Furthermore, after expanding CAR- $\gamma\delta$ T cells for an additional 5 days, we observed comparable lysis rates of target cells (Figure 4G), indicating the persistent CAR expression via endogenous promoter.

2.5 | CAR- $\gamma\delta$ T Cells Suppressed Tumor Growth in a Xenograft Mouse Model

Finally, we accessed the in vivo anti-tumor efficacy of our CAR- $\gamma\delta$ T cells using an ALL-mouse model. Although we knocked out TCR during the generation process of CAR- $\gamma\delta$ T cells, $\gamma\delta$ T cells could still utilize innate immune receptors such as NCR and NKG2D to mediate innate immune functions. Some studies have shown that CAR- $\gamma\delta$ T cells can exhibit significant cytotoxicity against tumor cells that are negative for tumor-associated antigens or lack MHC-I expression [26]. As shown in Figure 5A, mice were randomly divided into three groups: control group (untreated with PBS), $\gamma\delta$ T treatment group, and CAR- $\gamma\delta$ T treatment group. Equal numbers of $\gamma\delta$ T cells and CAR- $\gamma\delta$ T cells were transferred to NCG mice at 4 days and 11 days after infusion of Nalm6 cells, with IL2 administered three times weekly. We found that CAR- $\gamma\delta$ T cells successfully restrained the growth of Nalm6-cell-induced tumors in mice,

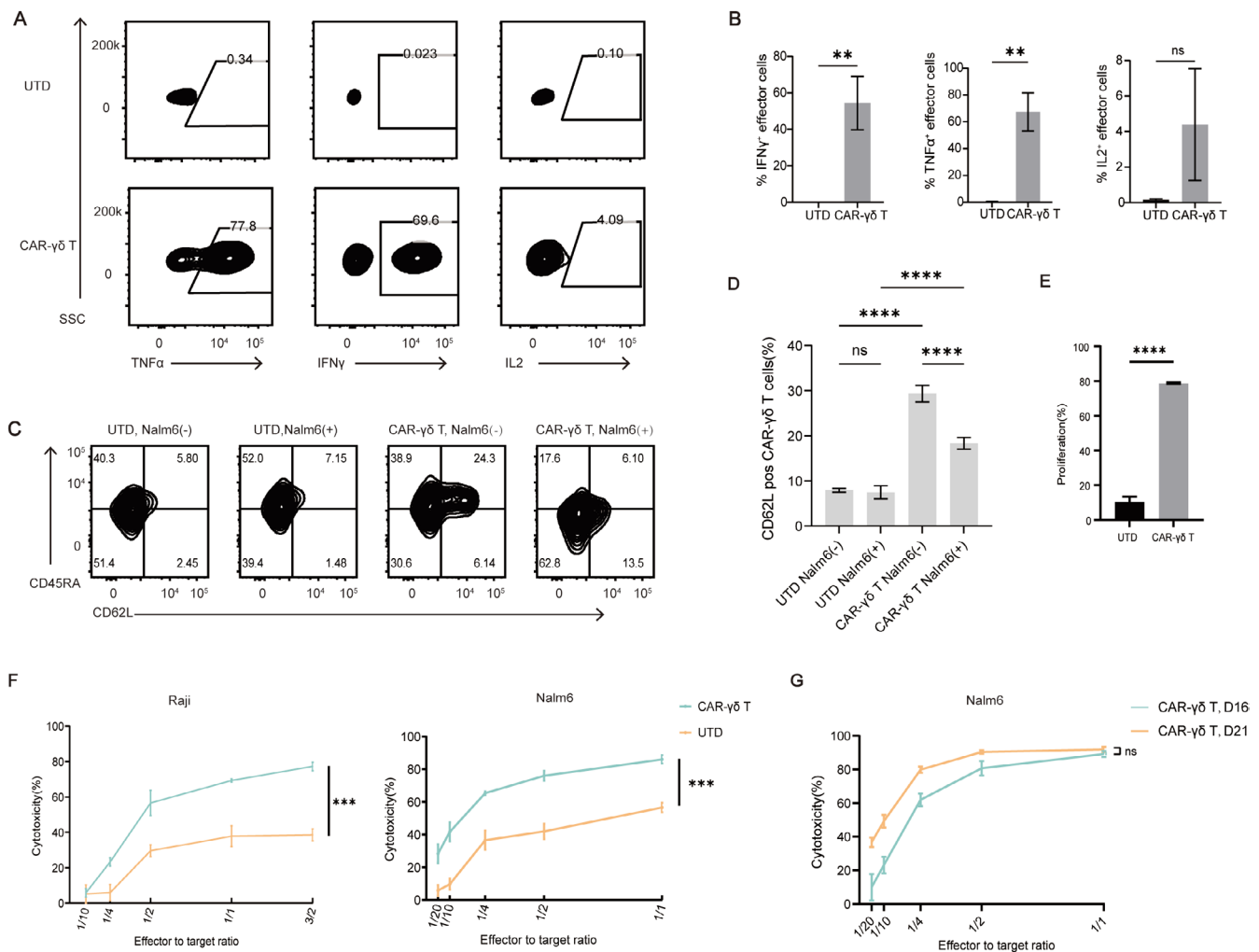


FIGURE 4 | The in vitro function of CAR- $\gamma\delta$ T cells. (A, B) Representative flow plots (A) and the FACS analysis (B) of cytokine release of CAR- $\gamma\delta$ T cells and untransduced $\gamma\delta$ T cells (UTD) upon stimulation with 2-fold irradiated CD19 positive Nalm6 cells for 18 h. Data are presented as mean $\pm$ SD ($n=3$, technical triplicates). (C, D) Representative flow plots (C) and the FACS analysis (D) of memory markers expression of CAR- $\gamma\delta$ T cells and UTD with/without stimulation of 2-fold irradiated CD19 positive Nalm6 cells for 7 days. (E) The proliferation of CFSE labeled CAR- $\gamma\delta$ T cells and UTD with/without stimulation of 2-fold irradiated CD19 positive Nalm6 cells for 4 days. The MFI values for unstimulated CAR- $\gamma\delta$ T cells or $\gamma\delta$ T cells utilized as a control. The percentage of proliferation was calculated as: (control MFI – experimental MFI)/(control MFI) $\times$ 100 (%). Data are presented as mean $\pm$ SD ($n=3$, technical triplicates). (F) Lysis using Nalm6 or Raji-FFLuc-GFP as target cells in 18 h cytotoxicity assay at different E:T ratios. (G) Lysis of CAR- $\gamma\delta$ T cells cultured at Day 16 (CAR- $\gamma\delta$ T, D16) and Day 21 (CAR- $\gamma\delta$ T, D21) targeting Nalm6-FFLuc-GFP. Lysis was determined as $(1 - (RLU_{sample}/RLU_{max})) \times 100$ (%) (RLU: relative luminescence units). Data are presented as mean $\pm$ SD ($n=3$, technical triplicates). p -values are determined by t -tests or ANOVA. ** $p < 0.01$; *** $p < 0.001$; **** $p < 0.0001$; ns, not significant ($p > 0.05$).

contributing to superior survival (Figure 5B,C). Interestingly, although UTD $\gamma\delta$ T cells showed partial anti-tumor function in vitro, they had little effect on tumor suppression in vivo (Figure 5B,C).

3 | Discussion

CAR- $\gamma\delta$ T cells have emerged as highly promising candidates in tumor immunotherapy recently, particularly due to their ability to circumvent GVHD and their cytotoxicity potential against broader kinds of tumors [27]. Currently, the manufacturing of CAR- $\gamma\delta$ T cells primarily relies on lentivirus infection [23, 28], which may introduce potential risks such as heterogeneity of CAR expression level [11], random gene-insertion [8] and transgene silencing in long term follow up [29]. Moreover,

transposon-based approaches (e.g., *Sleeping Beauty*, *Tc Buster*), which are applied in CAR- $\gamma\delta$ T cell preparation [30], have been reported to induce malignant transformation risks [31].

Recent advances in gene-editing techniques for $\alpha\beta$ T and NK cells have enabled the combination of Cas9/RNP complexes with AAV-delivered homology templates for site-specific integration into “safe harbor” loci (e.g., *AAVS1*, *CCR5*) [32], achieving >90% locus specificity [33]. In this study, we present an optimized methodology for CAR- $\gamma\delta$ T cell manufacturing, employing CRISPR/Cas9-mediated *TRDC* locus knockout combined with AAV-directed homologous recombination for site-specific integration of a CAR expression cassette. Our preliminary investigations revealed that conventional lentiviral transduction approaches resulted in a significant reduction in CAR expression levels within 7 days of in vitro culture (unpublished data).

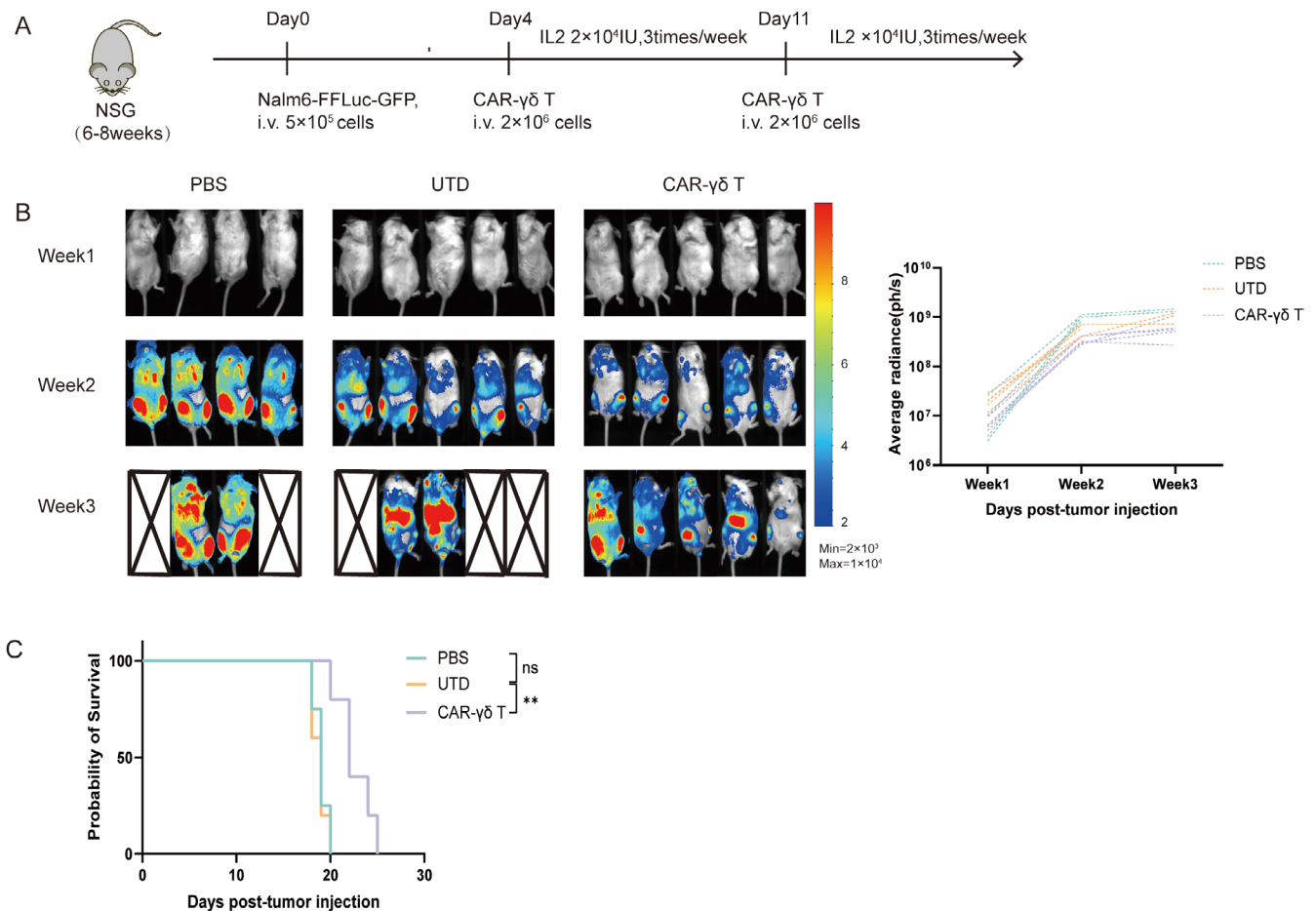


FIGURE 5 | CAR- $\gamma\delta$ T cell exhibited anti-tumor functionality in vivo. (A) Schematic of experimental protocol. NCG mice were injected intravenously with 5×10^5 Nalm6-FFLuc-GFP and administered intravenously with UTD cells or CAR- $\gamma\delta$ T cells at 4 and 11 days after Nalm6 inoculation, followed by thrice weekly intraperitoneal injections of 2×10^4 IU human recombinant IL2. (B) The bioluminescent imaging and quantification of leukemia burden (average radiance, ph/s) of mice treated with PBS, UTD, CAR- $\gamma\delta$ T cells, respectively ($n = 4$ mice for PBS group and $n = 5$ mice for other groups). (C) Kaplan-Meier analysis of the mice survival in each experimental group. ** indicates a statistically significant difference with $p < 0.01$.

The rationale for targeting the *TRDC* locus stems from the pioneering work in $\alpha\beta$ T cells, where CRISPR/Cas9-mediated CAR integration into the *TRAC* locus demonstrated superior therapeutic outcomes [11]. This approach, utilizing endogenous promoters rather than exogenous promoters, has been shown to maintain durable CAR stability, prevent tonic CAR signaling, delay T cell differentiation and exhaustion, and enhance therapeutic efficacy. Building upon these mechanistic insights, our study introduces an innovative strategy of targeting the *TRDC* locus in $\gamma\delta$ T cells for CAR gene integration, with the objective of achieving sustained and stable CAR expression through endogenous promoter regulation. However, it is crucial to consider that $\gamma\delta$ T cells possess intrinsic tumor recognition capabilities through their natural TCR, which can identify multiple tumor-associated antigens. The knockout of the *TRDC* locus potentially attenuate the inherent tumor recognition capacity of $\gamma\delta$ T cells to some degree.

Although the experimental data presented in Figures 3 and 4 demonstrate that CAR- $\gamma\delta$ T cells exhibit remarkable proliferative capacity and cytotoxic effects under in vitro culture conditions, the in vivo antitumor efficacy remains suboptimal and necessitates further refinement to enhance cytotoxic activity.

Notably, CAR-NK cells have been shown to achieve superior antitumor cytotoxicity by integrating cytoplasmic signaling domains of NK receptors (e.g., DAP10, 2B4, and DNAM-1) as modular signaling components [34, 35]. However, the CAR constructs employed in this study were primarily tailored for $\alpha\beta$ T cells and may not fully accommodate the distinct biological properties of $\gamma\delta$ T cells. Emerging evidence highlights the widespread expression of co-stimulatory molecules (e.g., CD27) in $\gamma\delta$ T cells, which play pivotal roles in modulating cell proliferation and cytokine production. Furthermore, a significant proportion of $\gamma\delta$ T cells express NK receptors such as NKG2D, whose signaling is predominantly mediated by DAP10 or DAP12 [5]. These receptors not only serve as co-stimulatory elements but also independently orchestrate antitumor responses. Therefore, incorporating these signaling domains into CAR constructs may better align with the intrinsic biology of $\gamma\delta$ T cells, thereby enhancing their activation efficiency and antitumor functionality.

Additionally, the tumor microenvironment in vivo is characterized by a significant upregulation of various inhibitory factors and their ligands including TGF- β and PD-L1 [36, 37]. Previous studies have confirmed that the PD-1/PD-L1 signaling pathway

can substantially suppress the proliferation and immune activity of $\gamma\delta$ T cells [38]. Thus, these inhibitory molecules may lead to a marked decline in cell performance and persistence of CAR- $\gamma\delta$ T cells in vivo.

A key consideration raised regarding our experimental design is the absence of a vblank AAV6-transduced control group to explicitly assess long-term vector backbone-associated safety in $\alpha\beta$ T cells. Although AAV-based gene therapy is widely recognized for its favorable safety profile in terms of immunotoxicity and genotoxicity in conventional $\alpha\beta$ T cell models [39], a critical knowledge gap exists: no safety studies of AAV in $\gamma\delta$ T cells have been documented to date. Accordingly, future study is warranted to systematically evaluate the safety profile of AAV, including AAV6, in $\gamma\delta$ T cells.

Notwithstanding certain limitations that warrant further refinement, this study first develops the site-specific genomic integration method in $\gamma\delta$ T cells, avoiding the risk of random gene integration. This protocol not only provides the optimal parameters for site-directed delivery of CAR gene in the *TRDC* locus of $\gamma\delta$ T cells but also lays the foundation for further gene-editing strategies for $\gamma\delta$ T cells. Future screening for “safe and efficient harbor” in $\gamma\delta$ T cells may be performed to explore better potential loci for gene KI.

4 | Methods and Materials

4.1 | Study Approval

All animal experiments were conducted and terminated in strict adherence to relevant guidelines and ethical regulations governing animal use, and were approved by the Ethics Committee of Hangzhou Normal University (Approval No.: HSD20210703). All experiments involving human material were approved in 2022 by the Clinical Research Ethics Committee of the First Affiliated Hospital, Zhejiang University School of Medicine (Approval No.: 1013-Quick). Healthy volunteers were recruited after informed written consent.

4.2 | Cell Lines and Culture Conditions

Raji, K562, and AAV-293 cells were purchased from the Cell Line Ontology Subset for Chinese National Infrastructure. Nalm6-FFluc-GFP was obtained from Shanghai Model Organisms. Raji-FFluc-GFP was generated by lentiviral vector transduction to label Raji cells with both green fluorescent protein (GFP) and luciferase (Luc). Adherent cell lines (AAV-293) and suspension cell lines (K562, Nalm6-FFluc-GFP, Raji-FFluc-GFP) were cultured in DMEM or RPMI 1640 supplemented with 10% fetal bovine serum (FBS) at 37°C and 5% CO₂. All cell lines were routinely tested negative for mycoplasma.

4.3 | Isolation and Expansion of Human $\gamma\delta$ T Cells

PBMCs were donated by healthy volunteers. PBMCs were isolated by density gradient centrifugation and cultured in X-vivo 15 medium (Lonza) supplemented with 10% FBS, 2 μ M

zoledronic acid (Novartis), and 250 U/mL recombinant human interleukin (IL)-2 (PeproTech). IL-7 and IL-15 (10 ng/mL each; PeproTech) were added as needed. The culture was maintained by semi-changing the medium every 3 days.

4.4 | Plasmid Construction and AAV Production

According to the previous studies, we designed a plasmid for CAR- $\gamma\delta$ T cells. Using pAAV-GFP (Addgene) as the backbone, we cloned the gRNA targeting sequences flanked by approximately 400 base pairs of the left homologous arm (LHA) and right homologous arm (RHA) of *TRDC*. A P2A peptide was inserted between LHA and RHA, followed by the gene cassette “1928z CAR-polyA.” All plasmids were prepared using standard molecular biology techniques and verified by Sanger sequencing.

The plasmids were transduced into AAV-293 cells along with the auxiliary pDP6 AAV Help/Cap/Rep replacement plasmid (Plasmid Factory) to produce the corresponding AAV6 virus according to the standard AAV production protocol.

4.5 | gRNA Design and Functional Screening

Base editor guide RNAs targeting *TRDC* sites were designed using the CRISPOR webtool (<http://crispor.tefor.net>). CRISPR/Cas9-mediated cleavage was conducted in K562 cells. Genomic DNA was extracted post-editing, with the *TRDC* target region amplified by PCR and Sanger-sequenced; Indel frequency was quantified using the online tool tracking of INDELS by decomposition (TIDE), a metric directly reflecting each sgRNA's capacity to mediate *TRDC* inactivation and enabling selection of high-efficiency gRNAs. The specific steps followed the protocol by Bak et al. [21].

4.6 | Genetic Modification of $\gamma\delta$ T Cells

PBMCs were induced in X-VIVO medium containing ZOL and IL2 for 5 and 10 days. The Cas9/ribonucleoprotein (RNP) complex consists of recombinant Cas9 protein and gRNA, enabling site-specific DNA cleavage at *TRDC* locus. A total of 240 pmol of gRNA (5'-TTCGTAGCCCCAAGCACTGTG-3', or alternative gRNA as needed, sequence details are provided in Figure S1A) (Genscript) was mixed with 10 μ g of Cas9 protein (or adjusted amounts as needed) (in-house produced), incubated at 37°C for 10 min to prepare the RNP. Approximately 2 $\times$ 10⁶ cells (or adjusted numbers as required) were centrifuged at 90 $\times$ g for 10 min, transferred to a 0.2 cm cuvette (ThermoFisher), and electroporated with Lonza IIB program X001 (or alternative programs T001, T023, U014 as needed) to deliver the RNP into $\gamma\delta$ T cells. Owing to issues pertaining to commercial confidentiality, the precise parameters of these programs remain undisclosed (e.g., the voltage and the duration of the program execution). After electroporation, the cells were resuspended in 1 mL of culture medium containing only 10% FBS and incubated for 0.5 h. The suspension was then centrifuged, the supernatant discarded, and the cells resuspended in complete culture medium at a concentration of 1 $\times$ 10⁷/mL for further culture. If subsequent AAV-mediated targeted insertion was required, AAV was added to

the cell culture medium at the appropriate MOI half an hour after electroporation.

4.7 | Cytotoxicity Assay

1×10^4 CAR- $\gamma\delta$ T cells and untransduced $\gamma\delta$ T cells were co-cultured with Nalm6 or Raji-FFLuc-GFP at various E:T ratios in a U-bottom deep well plate. Set up target cells of the same density in separate cultures as a control group to determine the maximum luciferase expression (relative light units; RLU). After 24 h, the cells were harvested and the Bright-Lite Luciferase Assay System (Vazyme) was used to quantify the target cells according to the standard protocol. Emission light was detected in a luminometer (Molecular Devices iD5). Lysis was calculated as $(1 - (\text{RLU sample})/(\text{RLU max})) \times 100$.

4.8 | Proliferation Assay

CAR- $\gamma\delta$ T cells and untransduced $\gamma\delta$ T cells were collected and centrifuged, then washed twice with PBS. They were re-suspended at a concentration of $1 \times 10^6/\text{mL}$ in PBS. CFSE (Invitrogen) was added to a final concentration of $5 \mu\text{M}$ and the mixture was incubated for 20 min at room temperature in the dark. Five times the volume of complete culture medium was then added to wash away excess dye. A total of 1×10^5 CAR- $\gamma\delta$ T cells/untransduced $\gamma\delta$ T cells were co-cultured with twice-irradiated Nalm6-FFLuc-GFP cells in a U-bottom 96-well tissue culture plates and incubated in a humidified incubator for 4 days. The cells were then harvested and subjected to FACS analysis.

4.9 | Phenotype and Exhaustion Assay

1×10^5 CAR- $\gamma\delta$ T cells and untransduced $\gamma\delta$ T cells were co-cultured with twice-irradiated Nalm6-FFLuc-GFP cells in a U-bottom 96-well tissue culture plates and incubated in a humidified incubator for 7 days. The cells were then harvested and subjected to FACS analysis.

4.10 | Cytokine Assay

1×10^5 CAR- $\gamma\delta$ T cells and untransduced $\gamma\delta$ T cells were co-cultured with twice-irradiated Nalm6-FFLuc-GFP cells in U-bottom 96-well tissue culture plates for 18 h. Golgi Plug protein transport inhibitor brefeldin A (BioLegend) was added at the beginning.

4.11 | Antibody, Extracellular and Intracellular Staining

Unless otherwise specified, fluorophore-conjugated antibodies were purchased from BioLegend. Extracellular staining antibodies included Alexa Fluor 647-Rabbit anti-mouse FMC63 scFv polyclonal antibody (BIOSWAN, clone: R19M, Cat#: 200102); PerCP/Cyanine5.5-CD45RA (clone: HI100, Cat#: 304122); Pacific Blue-CD62L (clone: DREG-56, Cat#: 304826);

APCCY7-PD1 (clone: EH12.2H7, Cat#: 329922); PE-LAG3 (clone: 11C3C65, Cat#: 369305); APCCY7-TIM3 (clone: F38-2E2, Cat#: 345026). Intracellular staining antibodies included Pacific Blue-IL2 (clone: MQ1-17H12, Cat#: 500324); PE-CY7-IFN γ (clone: 4S.B3, Cat#: 502528); PerCP/Cyanine5.5-TNF α (clone: MAb11, Cat#: 502926). Aqua (Cat#: 423102) was used as a viability dye, and FcR Blocking Reagent (Equitech Bio Inc.) was used to block Fc receptor staining. All staining was performed using optimized staining protocols provided by the antibody suppliers. Samples were acquired on LSRFortessa (BD Biosciences) or CytoFLEX and DxFLUX flow cytometers (BECKMAN COULTER), and analyzed using FlowJo v.10 software.

4.12 | Mouse Xenograft Model, CAR- $\gamma\delta$ T Treatment, and Tumor Monitoring

NOD/ShiLtJGpt-Prkdcem26Cd52Il2rgem26Cd22/Gpt (NCG) mice (female, aged 8 weeks) were purchased from Gempharmatech. Other parameters included: 12 h light/dark cycle, approximately 22°C, and 55% relative humidity.

The mice were randomly assigned to the control group (PBS), untransduced $\gamma\delta$ T cells treatment group, and CAR- $\gamma\delta$ T treatment group. Mice were injected via the tail vein with 5×10^5 Nalm6-FFLuc-GFP and then received intraperitoneal injections of PBS, 2×10^6 untransduced $\gamma\delta$ T cells and CAR- $\gamma\delta$ T cells according to their group assignment after 4 and 11 days, followed by thrice weekly intraperitoneal injections of 2×10^4 U human recombinant IL2. The thrice-weekly IL-2 dosing schedule was adopted based on preclinical evidence, our preliminary finding of enhanced $\gamma\delta$ T cell expansion and reduced exhaustion, and CAR- $\gamma\delta$ T cell therapy precedents (48-h intervals in mice, peri-infusion dosing) [23], to balance cytokine bioavailability and feasibility.

In vivo imaging of mice was performed weekly as follows. Each mouse received an intraperitoneal injection of 150 mg/kg D-luciferin and was anesthetized, followed by image acquisition 6 min later. The bioluminescence imaging device had an exposure time of 1 min for capturing images. Imaging data were analyzed using M3Vision. A consistent region of interest (ROI) was drawn over the tumor area, with average radiance measured as total photons per second (ph/s).

4.13 | Statistical Analysis

All experimental data are presented as the mean $\pm$ standard deviation (SD) and were analyzed using GraphPad Prism 9 software. Statistical comparisons were performed using *t*-tests or analysis of variance (ANOVA). For in vivo experiments, the overall survival comparison was determined by a Mantel-Cox log-rank test. *p*-values < 0.05 were statistically significant.

Author Contributions

Yingying Ding: methodology, investigation, formal analysis. **Ruoyu Dong:** conceptualization, writing – original draft, investigation, methodology, validation, software, formal analysis, data curation,

visualization, writing – review and editing. **Guangxun Yuan:** methodology, investigation, formal analysis, validation. **Haowen Xiao:** conceptualization, supervision, resources, writing – review and editing, methodology, project administration, funding acquisition. **Xiaoyan Yue:** methodology, investigation, formal analysis, validation. **Yixi Zhang:** methodology, data curation, validation, formal analysis, writing – review and editing, writing – original draft, conceptualization. **Xun Zeng:** conceptualization, methodology, supervision, project administration, resources.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data and materials supporting this study are available on reasonable request.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Figure S1:** cam471918-sup-0001-FigureS1.pdf.